基于三层组网数据直接转发的无线网络

一、实验目的

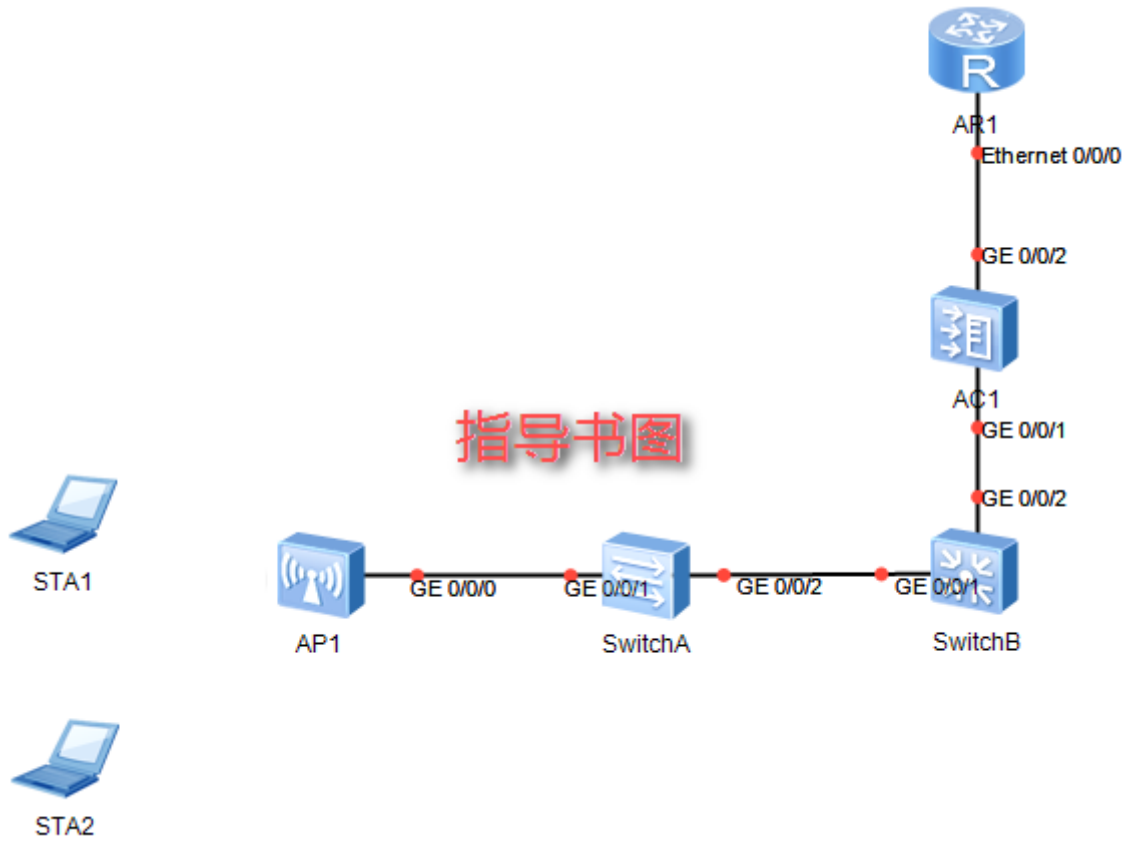
了解三层组网数据直接转发的无线网络相关知识，重点掌握三层组网的概念，配置方法。

二、模拟器版本

注意：需要 xxx 版本以上

三、实验拓扑

3.1 网络拓扑



3.2 参数规划表

配置项	规划数据
管理 VLAN	VLAN10、VLAN100
STA 业务 VLAN	VLAN pool
	名称：sta-pool
	VLAN pool 中加入的 VLAN：101、102
DHCP 服务器	AC 作为 AP 的 DHCP 服务器
	汇聚交换机作为 STA 的 DHCP 服务器
	STA 的默认网关为 10.23.101.2 和 10.23.102.2
AP 的 IP 地址池	10.23.100.2 ~ 10.23.100.254/24
STA 的 IP 地址池	10.23.101.3 ~ 10.23.101.254/24
	10.23.102.3 ~ 10.23.102.254/24
AC 的源接口 IP 地址	VLANIF100：10.23.100.1/24

AP 组	名称：ap1
	引用模板：VAP 模板 wlan-net、域管理模板 default
域管理模板	名称：default 国家码：中国
SSID 模板	名称：wlan-net
	SSID 名称：wlan-net
安全模板	名称：wlan-net 安全策略：WEP Open
VAP 模板	名称：wlan-net 转发模式：直接转发 业务 VLAN：VLAN pool

四、实验步骤

4.1 根据网络拓扑在模拟器中搭建网络；

4.2 步骤 1 配置周边设备

配置接入交换机SwitchA 的GE0/0/1 和GE0/0/2 接口加入VLAN10、VLAN101 和VLAN102，GE0/0/1 的缺省VLAN 为VLAN10。

```
<HUAWEI> system-view
[HUAWEI] sysname SwitchA
[SwitchA] vlan batch 10 101 102
[SwitchA] interface gigabitethernet 0/0/1
[SwitchA-GigabitEthernet0/0/1] port link-type trunk
[SwitchA-GigabitEthernet0/0/1] port trunk pvid vlan 10
[SwitchA-GigabitEthernet0/0/1] port trunk allow-pass vlan 10 101 102
[SwitchA-GigabitEthernet0/0/1] port-isolate enable
[SwitchA-GigabitEthernet0/0/1] quit
[SwitchA] interface gigabitethernet 0/0/2
[SwitchA-GigabitEthernet0/0/2] port link-type trunk
[SwitchA-GigabitEthernet0/0/2] port trunk allow-pass vlan 10 101 102
[SwitchA-GigabitEthernet0/0/2] quit
```

配置汇聚交换机SwitchB 的接口GE0/0/1 加入VLAN10、VLAN101 和VLAN102，接口GE0/0/2 加入VLAN100、VLAN101 和VLAN102，并创建接口VLANIF100，地址为10.23.100.2/24。

```
<HUAWEI> system-view
[HUAWEI] sysname SwitchB
[SwitchB] vlan batch 10 100 101 102
[SwitchB] interface gigabitethernet 0/0/1
[SwitchB-GigabitEthernet0/0/1] port link-type trunk
[SwitchB-GigabitEthernet0/0/1] port trunk allow-pass vlan 10 101 102
[SwitchB-GigabitEthernet0/0/1] quit
[SwitchB] interface gigabitethernet 0/0/2
[SwitchB-GigabitEthernet0/0/2] port link-type trunk
[SwitchB-GigabitEthernet0/0/2] port trunk allow-pass vlan 100 101 102
[SwitchB-GigabitEthernet0/0/2] quit
[SwitchB] interface vlanif 100
[SwitchB-Vlanif100] ip address 10.23.100.2 24
[SwitchB-Vlanif100] quit
```

配置Router 的接口GE1/0/0 加入VLAN101 和VLAN102，创建接口VLANIF101 并配置IP 地址为10.23.101.2/24，创建接口VLANIF102 并配置IP 地址为10.23.102.2/24。

```
<Huawei> system-view
[Huawei] sysname Router
[Router] vlan batch 101 102
[Router] interface gigabitethernet 1/0/0
[Router-GigabitEthernet1/0/0] port link-type trunk
[Router-GigabitEthernet1/0/0] port trunk allow-pass vlan 101 102
[Router-GigabitEthernet1/0/0] quit
[Router] interface vlanif 101
[Router-Vlanif101] ip address 10.23.101.2 24
[Router-Vlanif101] quit
[Router] interface vlanif 102
[Router-Vlanif102] ip address 10.23.102.2 24
[Router-Vlanif102] quit
```

4.3 步骤 2 配置 AC 与其它网络设备互通

配置AC 的接口GE0/0/1 加入VLAN100、VLAN101 和VLAN102，GE0/0/2 加入VLAN101 和VLAN102，并创建接口VLANIF100。

```
<AC6605> system-view
[AC6605] sysname AC
[AC] vlan batch 100 101 102
[AC] interface vlanif 100
[AC-Vlanif100] ip address 10.23.100.1 24
[AC-Vlanif100] quit
[AC] interface gigabitethernet 0/0/1
[AC-GigabitEthernet0/0/1] port link-type trunk
[AC-GigabitEthernet0/0/1] port trunk allow-pass vlan 100 101 102
[AC-GigabitEthernet0/0/1] quit
[AC] interface gigabitethernet 0/0/2
[AC-GigabitEthernet0/0/2] port link-type trunk
[AC-GigabitEthernet0/0/2] port trunk allow-pass vlan 101 102
[AC-GigabitEthernet0/0/2] quit
```

配置AC 到AP 的路由，下一跳为SwitchB 的VLANIF100。

```
[AC] ip route-static 10.23.10.0 24 10.23.100.2
```

4.4 步骤 3 配置 **DHCP** 服务器为 **STA** 和 **AP** 分配 **IP** 地址

在SwitchB 上配置DHCP 中继，代理AC 分配IP 地址。

```
[SwitchB] dhcp enable
[SwitchB]ip pool vlan101
[SwitchB-ip-pool-vlan101]gateway-list 10.23.101.2
[SwitchB-ip-pool-vlan101]network 10.23.101.0 mask 255.255.255.0
[SwitchB-ip-pool-vlan101]quit
[SwitchB]ip pool vlan102
[SwitchB-ip-pool-vlan102]gateway-list 10.23.102.2
[SwitchB-ip-pool-vlan102]network 10.23.102.0 mask 255.255.255.0
[SwitchB-ip-pool-vlan102]quit
[SwitchB] interface vlanif 10
[SwitchB-Vlanif10] ip address 10.23.10.1 24
[SwitchB-Vlanif10] dhcp server option 43 sub-option 3 ascii 10.23.100.1
[SwitchB-Vlanif10] quit
```

在SwitchB 上创建VLANIF101 和VLANIF102 接口为STA 提供地址，并指定默认网关。

```
[SwitchB] interface vlanif 101
[SwitchB-Vlanif101] ip address 10.23.101.1 24
[SwitchB-Vlanif101] dhcp select global
[SwitchB-Vlanif101] quit
[SwitchB] interface vlanif 102
[SwitchB-Vlanif102] ip address 10.23.102.1 24
[SwitchB-Vlanif102] dhcp select global
[SwitchB-Vlanif102] quit
```

4.5 步骤 4 配置 VLAN pool，用于作为业务 VLAN

在AC 上新建VLAN pool，并将VLAN101 和VLAN102 加入其中，配置VLAN pool 中的VLAN 分配算法为“hash”。

```
[AC] vlan pool sta-pool
[AC-vlan-pool-sta-pool] vlan 101 102
[AC-vlan-pool-sta-pool] assignment hash
[AC-vlan-pool-sta-pool] quit
```

4.6 步骤 5 配置 AP 上线

创建AP 组，用于将相同配置的AP 都加入同一AP 组中。

```
[AC] wlan
[AC-wlan-view] ap-group name ap1
[AC-wlan-ap-group-ap1] quit
```

创建域管理模板，在域管理模板下配置AC 的国家码并在AP 组下引用域管理模板。

```
[AC-wlan-view] regulatory-domain-profile name default
[AC-wlan-regulate-domain-default] country-code cn
[AC-wlan-regulate-domain-default] quit
[AC-wlan-view] ap-group name ap1
[AC-wlan-ap-group-ap1] regulatory-domain-profile default
Warning: Modifying the country code will clear channel, power and antenna gain configurations
of the radio and reset the AP. Continue?[Y/N]:y
[AC-wlan-ap-group-ap1] quit
[AC-wlan-view] quit
```

配置AC 的源接口。

```
[AC] capwap source interface vlanif 100
```

在AC 上离线导入AP，并将AP 加入AP 组“ap1”中

```
[AC] wlan
[AC-wlan-view] ap auth-mode mac-auth
[AC-wlan-view] ap-id 0 ap-mac 00e0-fc6e-1310
[AC-wlan-ap-0] ap-name area_1
[AC-wlan-ap-0] ap-group ap1
Warning: This operation may cause AP reset. If the country code changes, it will clear channel,
power and antenna gain configurations of the radio, Whether to continue? [Y/N]:y
[AC-wlan-ap-0] quit
```

将AP 上电后，当执行命令display ap all 查看到AP 的“State”字段为“nor”时，表示AP 正常上线。

```
<AC>display ap all
Info: This operation may take a few seconds. Please wait for a moment.done.
Total AP information:
nor  : normal          [1]
-------------------------------------------------------------------------------
---------
ID   MAC            Name    Group IP          Type            State STA Uptime
-------------------------------------------------------------------------------
---------
0    00e0-fc6e-1310 area_1 ap1   10.23.10.254 AP4050DN-E      nor   0   9S
-------------------------------------------------------------------------------
---------
Total: 1
<AC>
```

指导书图

4.7 步骤 6 配置 WLAN 业务参数

创建名为“wlan-net”的安全模板，并配置安全策略。

[AC-wlan-view] security-profile name wlan-net

[AC-wlan-sec-prof-wlan-net] security open

[AC-wlan-sec-prof-wlan-net] quit

创建名为“wlan-net”的SSID 模板，并配置SSID 名称为“wlan-net”。

[AC-wlan-view] ssid-profile name wlan-net

[AC-wlan-ssid-prof-wlan-net] ssid wlan-net

[AC-wlan-ssid-prof-wlan-net] quit

创建名为“wlan-net”的VAP 模板，配置业务数据转发模式、业务VLAN，并且引用安全模板和SSID 模板。

[AC-wlan-view] vap-profile name wlan-net

[AC-wlan-vap-prof-wlan-net] forward-mode direct-forward

[AC-wlan-vap-prof-wlan-net] service-vlan vlan-pool sta-pool

[AC-wlan-vap-prof-wlan-net] security-profile wlan-net

[AC-wlan-vap-prof-wlan-net] ssid-profile wlan-net

[AC-wlan-vap-prof-wlan-net] quit

配置AP 组引用VAP 模板，AP 上射频0 和射频1 都使用VAP 模板“wlan-net”的配置。

[AC-wlan-view] ap-group name ap1

[AC-wlan-ap-group-ap1] vap-profile wlan-net wlan 1 radio 0

[AC-wlan-ap-group-ap1] vap-profile wlan-net wlan 1 radio 1

[AC-wlan-ap-group-ap1] quit

4.8 测试网络

验证配置结果

WLAN 业务配置会自动下发给AP，配置完成后，通过执行命令display vap ssid wlan-net 查看如下信息，当“Status”项显示为“ON”时，表示 AP 对应的射频上的 VAP 已创建成功。

```
Total: 1
<AC>display vap ssid wlan-net
Info: This operation may take a few seconds, please wait.
WID : WLAN ID
--------------------------------------------------------------------------
AP ID AP name RfID WID  BSSID            Status  Auth type  STA   SSID
--------------------------------------------------------------------------
0     area_1  0    1    00E0-FC6E-1310 ON      Open       0     wlan-net
0     area_1  1    1    00E0-FC6E-1320 ON      Open       0     wlan-net
--------------------------------------------------------------------------
Total: 2
<AC>
```

#STA 搜索到名为“wlan-net”的无线网络，组网采用Open-system 认证，用户不需要输入密码。用户关联到无线网络上后，无线PC 能够被分配相应的IP 地址。在AC 上执行display station ssid wlan-net 命令，可以查看到用户已经接入到无线网络“wlan-net”中。

```
<AC>display station ssid wlan-net
Rf/WLAN: Radio ID/WLAN ID
Rx/Tx: link receive rate/link transmit rate(Mbps)
--------------------------------------------------------------------------------
-----------
STA MAC          AP ID Ap name  Rf/WLAN  Band  Type  Rx/Tx      RSSI  VLAN  IP a
ddress
--------------------------------------------------------------------------------
-----------
5489-981c-515d   0     area_1   0/1      2.4G  -     -/-        -     102   10.2
3.102.254
--------------------------------------------------------------------------------
-----------
Total: 1 2.4G: 1 5G: 0
<AC>
```

#sta 的 ip 地址为路由器的 ip

STA1

Vap 列表 | 命令行 | UDP发包工具

```
Welcome to use STA Simulator!

STA>ipconfig

Link local IPv6 address...........: ::
IPv6 address......................: / 128
IPv6 gateway......................:
IPv4 address......................: 10.23.102.254
Subnet mask.......................: 255.255.255.0
Gateway...........................: 10.23.102.2
Physical address..................: 54-89-98-1C-51-5D
DNS server........................:

STA>
```

指导书图